SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA0706 COURSE NAME: COMPUTER NETWORKS

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I B.Pardhu Arjun (192525245) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
B.Pardhu Arjun

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X3+1.

   i.) Use polynomial long division to determine the message that should be transmitted.

   ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

5. A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps), home Wi-Fi (20 Mbps), and mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1 — CRC Error Detection

Message: 11100011 Generator polynomial: $X^3 + 1$ → bit pattern 1001

(i) Polynomial Long Division — Transmitted Message

Since the generator has degree 3, append 3 zero bits to the message before dividing:

```
Augmented message : 11100011000
Divisor (1001)    : 1001
```

We perform modulo-2 (XOR) division of the augmented message 11100011000 by the divisor 1001. At each step, the divisor is XORed under the current leftmost 1-bit, which cancels that bit to 0; we then move to the next leftmost 1 and repeat, until fewer than 4 bits remain — those final bits form the remainder.

```
Row 0 (dividend):  1 1 1 0 0 0 1 1 0 0 0 XOR 1001 (aligned at b1) :  1 0 0 1 Row 1
:  0 1 1 1 0 0 1 1 0 0 0  XOR 1001 (aligned at b2) :    1 0 0 1 Row 2
:  0 0 1 1 1 0 1 1 0 0 0  XOR 1001 (aligned at b3) :      1 0 0 1 Row 3
:  0 0 0 1 1 1 1 1 0 0 0  XOR 1001 (aligned at b4) :        1 0 0 1 Row 4
:  0 0 0 0 1 1 0 1 0 0 0  XOR 1001 (aligned at b5) :          1 0 0 1 Row 5
:  0 0 0 0 0 1 0 0 0 0 0  XOR 1001 (aligned at b6) :            1 0 0 1 Row 6
:  0 0 0 0 0 0 0 0 1 0 0  Only 3 bits remain after the last leading 1 (positions
9,10,11) - too few to align the 4-bit divisor again, so division stops here.  Remainder
= last 3 bits of Row 6 = 1 0 0
```

Remainder (CRC) = 100

Verification: dividing the full codeword 11100011100 (message + remainder) by 1001 using the same process gives a remainder of 000, confirming the remainder was computed correctly.

Transmitted codeword = Message + Remainder = 11100011 + 100 = 11100011100

(ii) Effect of Inverting the Leftmost Bit

The leftmost bit of the transmitted codeword corresponds to the highest-order bit (position x^{10}). Flipping this bit is equivalent to adding an error polynomial $E(x) = x^{10}$ to the transmitted codeword $T(x)$.

Corrupted codeword: 01100011100 (leftmost 1 flipped to 0)

At the receiver, the corrupted codeword 01100011100 is divided again by the generator (1001), using the same long-division method as in part (i):

```
Row 0 (received)         :  0 1 1 0 0 0 1 1 1 0 0 XOR 1001 (aligned at b2) :    1 0 0 1
Row 1                    :  0 0 1 1 0 0 1 1 1 0 0  XOR 1001 (aligned at b3) :      1 0
0 1 Row 2                    :  0 0 0 1 1 1 1 1 1 0 0  XOR 1001 (aligned at b4) :
1 0 0 1 Row 3                    :  0 0 0 0 1 0 1 1 1 0 0  XOR 1001 (aligned at b5) :
1 0 0 1 Row 4                    :  0 0 0 0 0 1 1 0 1 0 0  XOR 1001 (aligned at b6) :
1 0 0 1 Row 5                    :  0 0 0 0 0 0 1 1 1 0 0  XOR 1001 (aligned at b7) :
1 0 0 1 Row 6                    :  0 0 0 0 0 0 0 1 0 0 0  Remainder = last 3 bits of
Row 6 = 0 1 0
```

Remainder at receiver = 010 (non-zero)

This can also be confirmed algebraically: flipping the leftmost bit corresponds to adding an error polynomial $E(x) = x^{10}$. Since the original codeword was exactly divisible by $G(x)$, the new remainder equals $E(x) \bmod G(x)$. Using $x^3 \equiv 1 \pmod{G(x)}$, we get $x^{10} = (x^3)^3 \cdot x \equiv x$, i.e. remainder = 010 — matching the long-division result above.

Conclusion: Because an error-free codeword always divides G(x) exactly (remainder = 000), a non-zero remainder at the receiver is the signal that a transmission error has occurred. The receiver detects the error, discards the frame, and (depending on protocol) requests retransmission.

Question 2 — Throughput & Channel Capacity

(a) Throughput Calculation

Given: Bandwidth = 10 Mbps, Frames = 12,000 per minute, Bits per frame = 10,000

```
Throughput = (Frames/min × Bits/frame) / 60 sec
           = (12,000 × 10,000) / 60
           = 120,000,000 / 60
           = 2,000,000 bps = 2 Mbps
```

Interpretation: Although the network's bandwidth (capacity) is 10 Mbps, the actual achieved throughput is only 2 Mbps — this gap illustrates why throughput (measured performance) and bandwidth (theoretical capacity) are distinct networking metrics.

(b) Channel Capacity — Shannon's Formula

Given: Bandwidth B = 3000 Hz, Signal-to-Noise Ratio (SNR) = 3162

```
C = B × log2(1 + SNR)
  = 3000 × log2(1 + 3162)
  = 3000 × log2(3163)
  ≈ 3000 × 11.63
  ≈ 34,860 bps ≈ 34.86 kbps
```

Interpretation: This is the theoretical maximum error-free data rate achievable on this noisy 3000 Hz channel.

Question 3 — Hamming (12,8) Code for Hospital Patient Monitoring

Scenario: Wireless patient-monitoring devices transmit heart rate and blood pressure data. Occasional electromagnetic interference flips a single bit. The hospital uses Hamming (12,8) so that a single-bit error can be corrected instantly at the receiver, without requiring retransmission.

Code Structure

With m = 8 data bits, the number of redundancy (parity) bits r must satisfy: 2^r ≥ m + r + 1. r = 4 satisfies this (2^4 = 16 ≥ 13), giving a total codeword length n = m + r = 12 bits — hence Hamming (12,8).

Position	1	2	3	4	5	6	7	8	9	10	11	12
Bit Type	P1	P2	D1	P4	D2	D3	D4	P8	D5	D6	D7	D8

Parity bits are placed at power-of-2 positions (1, 2, 4, 8); data bits fill the remaining positions.

Parity Bit Coverage

- P1 (position 1): covers all positions whose binary index has bit-0 set → 1, 3, 5, 7, 9, 11
- P2 (position 2): covers all positions whose binary index has bit-1 set → 2, 3, 6, 7, 10, 11
- P4 (position 4): covers all positions whose binary index has bit-2 set → 4, 5, 6, 7, 12
- P8 (position 8): covers all positions whose binary index has bit-3 set → 8, 9, 10, 11, 12

Each parity bit is set (0 or 1) so that the total number of 1s in its coverage group is even (even parity).

Example Encoding

Let the 8-bit patient data be: 1 0 1 1 0 1 0 1, placed into D1–D8 respectively. Each parity bit is computed as the XOR of the data bits it covers, ensuring even parity in each group. This produces the final 12-bit codeword that is transmitted over the wireless channel.

Error Detection and Correction at the Receiver

When the 12-bit codeword arrives, the receiver recomputes each parity check and XORs the received parity bit with the recalculated value, forming a 4-bit binary syndrome (S8 S4 S2 S1):

- If the syndrome = 0000 → no error occurred; data accepted as-is.
- If the syndrome is non-zero → its decimal value directly gives the exact bit position that is in error (this is the key property of Hamming codes).
- The receiver flips that exact bit, correcting the single-bit error instantly.

Relevance to the Hospital Scenario

This single-bit error-correcting capability is exactly what the hospital needs: since electromagnetic interference typically corrupts only one bit at a time, Hamming (12,8) lets the monitoring station correct the error in real time and deliver accurate heart-rate/blood-pressure readings to doctors immediately — without the delay of detecting an error and requesting retransmission, which is critical in continuous patient-monitoring applications.

Question 4 — IPv4 Subnetting and IPv6 Addressing

(a) IPv4 Subnetting: 172.16.10.0/24 → /27

Bits borrowed for subnetting = 27 − 24 = 3 → Total subnets = 2^3 = 8

Host bits remaining = 32 − 27 = 5 → Usable hosts per subnet = $2^5 - 2$ = 30

Subnet mask = 255.255.255.224 | Block size (increment) = 32

Subnet #	Network Address	Usable Host Range	Broadcast Address
1	172.16.10.0	172.16.10.1 – 172.16.10.30	172.16.10.31
2	172.16.10.32	172.16.10.33 – 172.16.10.62	172.16.10.63
3	172.16.10.64	172.16.10.65 – 172.16.10.94	172.16.10.95
4	172.16.10.96	172.16.10.97 – 172.16.10.126	172.16.10.127
5	172.16.10.128	172.16.10.129 – 172.16.10.158	172.16.10.159
6	172.16.10.160	172.16.10.161 – 172.16.10.190	172.16.10.191
7	172.16.10.192	172.16.10.193 – 172.16.10.222	172.16.10.223
8	172.16.10.224	172.16.10.225 – 172.16.10.254	172.16.10.255

The IP address 172.16.10.78 falls within Subnet 3 (range 64–95):

- Network address: 172.16.10.64
- Broadcast address: 172.16.10.95

- Usable host range: 172.16.10.65 – 172.16.10.94

Checking 172.16.10.95: Yes, it lies within the same subnet range (64–95). However, it is the broadcast address of Subnet 3, not an assignable host address.

(b) IPv6 Addressing

Given address: 2001:0db8:0000:0000:0000:ff00:0042:8329

Compressed form (drop leading zeros in each group; replace the longest run of consecutive all-zero groups with ::):

```
2001:db8::ff00:42:8329
```

Expanded (full) form (restore every group to 4 hex digits):

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

Network prefix using /64 (first 64 bits = first four groups):

```
2001:0db8:0000:0000::/64    (or equivalently 2001:db8::/64)
```

Total possible host addresses = remaining bits = 128 − 64 = 64 bits:

```
Total host addresses = 2^64
```

Question 5 — Online Exam Server: Flow Control & Congestion Scenario

Primary Networking Problem

The scenario describes a classic Flow Control mismatch between sender and receiver, which in turn triggers network-level Congestion. The exam server (a very fast 10 Gbps sender) transmits data faster than many student devices (slow, heterogeneous 5–100 Mbps receivers) can accept and process it.

How the Speed Mismatch Affects Communication

- Receiver Buffer Overflow: Each student device/router has a finite receive buffer. If the server keeps sending faster than the buffer is drained by the application, the buffer fills up and additional incoming packets are dropped.
- Unnecessary Retransmissions: Dropped packets cause timeouts or duplicate-ACK detection at the sender, forcing retransmission of the same data — which adds further load to an already strained network, creating a feedback loop that worsens congestion.
- Delayed Acknowledgments: Overloaded receivers cannot generate ACKs promptly, stalling the sender's transmission window and increasing perceived latency — this is experienced by students as a frozen exam portal.
- Increased Network Traffic / Congestion: With 8,000 students connecting simultaneously, the aggregate effect of retransmissions and duplicate packets consumes significant shared bandwidth, amplifying congestion beyond the individual flow-control problem at each receiver.

How This Should Be Resolved

Protocols such as TCP address exactly this problem using a sliding-window flow-control mechanism: the receiver advertises a receive window (rwnd) that tells the sender how much unacknowledged data it can currently accept, and the sender must throttle its rate to stay within that window.

In addition, congestion-control algorithms (e.g., TCP Slow Start and Congestion Avoidance) allow the server to ramp up its transmission rate gradually per connection instead of transmitting at full capacity immediately — reducing buffer overflow, retransmission storms, and portal freezing for students on slower connections.